

Gauge equivalence does not survive quantisation: value-output orbit, norm-product bound, and consequences for rotational methods

Manuel Muñoz Plá

Independent researcher — Seville, Spain

ORCID: 0009-0000-5714-912X mmunozpl@uoc.edu

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Abstract

Rotational quantisation methods exploit a computational invariance of attention’s value-output sector: rotating the pair $(\mathbf{W}_v, \mathbf{W}_O)$ without altering the function. That invariance is a subgroup of a larger orbit — $GL(d_h)$ per head, certified in [1]— and practice restricts it twice: to orthogonal rotations, and shared across the heads of a layer. Here we measure what both restrictions buy in terms of the OV circuit’s quantisation error, on two architectures (Pythia-410M and a fine-tuned ViT-B/16), with a deliberately simple quantiser that isolates the variable of interest. Three results. Within the orthogonal subgroup the orbit is flat: the error range across rotations stays at $\times 1.005$ – 1.02 in median, and the trained gauge is already within $\times 1.02$ – 1.04 of the best sampled point. Per-head freedom improves on the shared rotation by 1.3 per cent, below the threshold the design fixed as material. And in the sampled GL family, increasing anisotropy raises the error monotonically, up to $\times 15$ in median and $\times 10^5$ at the extreme. The mechanism admits a closed form: the error is bounded by the product of the transformed factors’ norms, and the approximation that follows from the bound predicts exponent one, with 0.998 measured across two gauge families, so that the catastrophic tail turns out to be the dispersion of conditioning transmitted by that relation. The bound also names the design space’s border, which does not separate orthogonal from non-orthogonal: the conformal orthogonal class leaves the bound’s factor p invariant by identity, the balanced point —non-orthogonal— minimises the bound over the entire orbit, and the trained gauge sits within 1.1–1.5 per cent of that optimum. An end-to-end validation confirms the flatness in real accuracy: identity, best and worst sampled rotation and balanced point fall within 0.10 percentage points, indistinguishable by paired McNemar over the same 5000 images. The decisive contrast is a different one. A general gauge at scale 2, unquantised, predicts the same class as the identity on all 5000 images, without a single discrepancy. Quantised to int4 it falls to 0.0082, below nominal chance over 100 classes, and then agrees with the identity on 44 images. The control isolates the cause: gauge equivalence is exact in real arithmetic and does not survive the grid. The result further delimits where the gain of rotational methods does not lie —not in the orbit’s weight landscape—, consistent with their usual attribution to activation outliers.

Keywords: quantisation, computational invariance, gauge orbit, OV circuit, multi-head attention, RTN.

1 Motivation

A gauge freedom runs through the value-output sector of every attention head: the substitution $\mathbf{W}_v \rightarrow \mathbf{W}_v \mathbf{R}$, $\mathbf{W}_O \rightarrow \mathbf{R}^{-1} \mathbf{W}_O$, with $\mathbf{R}$ invertible in $GL(d_h)$, leaves the layer’s output and the circuit $\text{OV} = \mathbf{W}_O \mathbf{W}_v$ exactly intact. For readability the value bias is omitted; when present it transforms together with the value projection, which in the storage convention of section 2 reads $\mathbf{b}_v \mapsto \mathbf{R}^\top \mathbf{b}_v$. Omitting it does not leave the output intact, and section 3.5 measures that. In [1]

that orbit is certified per head and its consequence for geometric readings of $\mathbf{W}_O$ is measured. Here its other side is of interest. Rotational quantisation —QuaRot [2] with fixed Hadamard rotations, SpinQuant [3] with learned rotations— exploits precisely this invariance to redistribute the weights before quantising them: its rotation R_2 operates on the value-output pair, restricted to orthogonal matrices and shared across all heads of the layer.

Practice, therefore, picks a point of the orbit by imposing two restrictions —orthogonality and sharing— with no characterisation of what not imposing them would cost, nor of how much is left on the table within them. Here we measure both with the formal framework in hand. Three concrete questions, with their reading rules fixed before seeing any data: **Q1**, how much does the circuit’s quantisation error vary along the orthogonal orbit? **Q2**, where does the trained gauge —the weights as the optimiser left them— sit relative to that range? **Q3**, what does per-head freedom buy over the shared rotation practice uses?

A word on the kind of work this is, before starting: a characterisation of the space, not a method. There is no GPTQ, no learned rotations, no comparison against the quantisation state of the art. The quantiser is deliberately the simplest possible because the object being measured is the gauge, not the quantiser.

2 Experimental design

Models and objects. Two architectures with the same $d_h = 64$: Pythia-410M (24 layers, 16 heads; public weights, no training of our own) and a ViT-B/16 fine-tuned on ImageNet-100 (12 layers, 12 heads; seed 42 from [1]). One realisation per model: the dispersion of interest is across gauges, heads, and layers, not across seeds, and this is declared as such. The pair $(\mathbf{W}_v^{(h)}, \mathbf{W}_O^{(h)})$ per head is quantised. With the storage orientation of linear layers, $\mathbf{v} = \mathbf{x}\mathbf{W}_v^\top + \mathbf{b}_v$, the gauge of section 1 reads $\mathbf{W}_v \mapsto \mathbf{R}^\top \mathbf{W}_v$ and $\mathbf{W}_O \mapsto \mathbf{W}_O \mathbf{R}^{-\top}$, which is the form used from section 3.4 onwards. The reference is the exact circuit $\mathbf{M}_h = \mathbf{W}_O^{(h)} \mathbf{W}_v^{(h)}$ in fp64 and the metric the relative Frobenius error $e = \|\hat{\mathbf{M}} - \mathbf{M}\|_F / \|\mathbf{M}\|_F$, with $\hat{\mathbf{M}}$ recomposed in fp64. All gauge arithmetic is in fp64: with a strong $\mathbf{R}$, the inverse in single precision amplifies the error by the condition number and contaminates the measurement —the phenomenon that section 3.4 turns into a result.

Quantiser and regimes. Symmetric per-channel (row) RTN, at int4 and int8, with no data calibration. Per layer and head: identity (the weights as they are); per-head orthogonal ($K = 32$ independent Haar rotations); shared orthogonal (the same $K = 32$ across every head of the layer, SpinQuant’s regime); and general GL ($\mathbf{R} = \text{noise} + \text{scale} \cdot \mathbf{I}$, $\text{scale} \in \{2, 8, 32\}$, $K = 32$ per scale — low scale denotes a strong gauge and a high condition number). 170 016 measurements in total (123 648 on Pythia, 46 368 on ViT-B), with exact counts and zero duplicate keys.

Prior gates. Three, and the sweep does not run without them: the orthogonal gauge without quantising reproduces the circuit at the double-precision floor (worst case 1.9×10^{-15}); int8 on the identity gives an error one order of magnitude smaller than int4 (ratio $\approx 18\times$ on both models); and the quantiser is exactly idempotent.

Reading rules, fixed before the data. Q1: the orthogonal range is “material” if the worst/identity ratio reaches $\times 2$, “narrow” if it stays under $\times 1.25$. Q2: the identity is “far from the floor” if it exceeds $\times 1.5$ the best sampled point, “close” if it stays under $\times 1.1$. Q3: per-head improvement is “material” if its median reaches 20 % relative, “null” if it stays under 5 %. Every pre-specified outcome is reported, whichever way it falls.

A fourth rule, that of section 3.5, was fixed *after* seeing Q1 —an explicit update, not a silent deviation—. The original preregistration called for sign coherence (best $\geq$ identity $\geq$ worst) across three forwards. Q1 showed those three points differ in e by only 2–8 %, a distance three

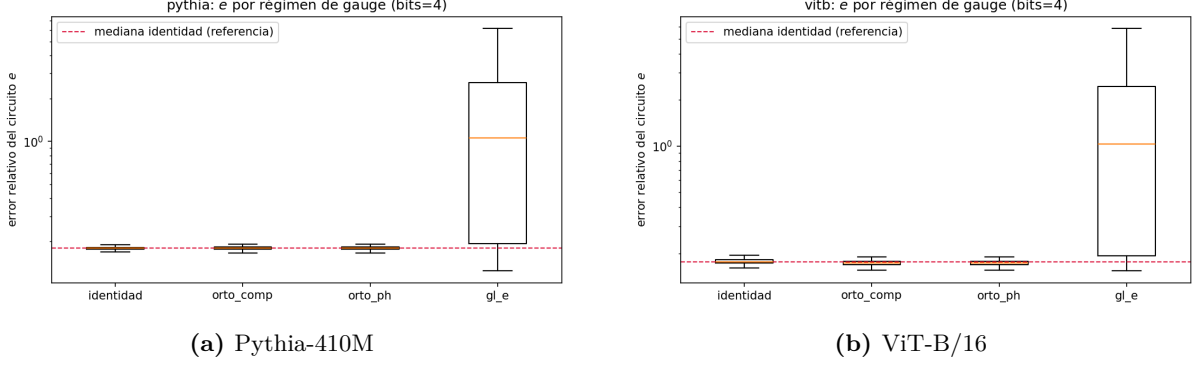


Figure 1: Distribution of the circuit’s relative error e per gauge regime (bits=4), with the identity’s median as a horizontal reference. The outlier-free boxes show the flatness between the identity and both orthogonal regimes.

forwards cannot resolve as a sign without confusing noise with signal. The updated rule, fixed before running the forwards: if the orthogonal range is $< \times 1.25$ (it is), the prediction becomes indistinguishability, and that indistinguishability, if confirmed, is the confirmation of Q1 in real accuracy. It compares the three points *against each other* by range: it neither predicts nor rules out a systematic difference between the identity and the orthogonal set, which if it appeared would be a different, unregistered observation. Declared post-hoc forwards were also added, marked as such in the table: a general gauge at scale 2, with and without quantisation, to anchor the toxic tail in an accuracy figure and isolate its cause, and the balanced point of corollary 2.

3 Results

3.1 The orthogonal orbit is flat (Q1)

Per cell (layer, head, width), the ratio between the worst sampled orthogonal error and the identity’s error lands on “narrow” across the totality of cells in both models: 768 of 768 on Pythia (median $\times 1.02$, range $[0.97, 1.08]$) and 288 of 288 on ViT-B (median $\times 1.005$, range $[0.76, 1.06]$). Picking one or another orthogonal rotation barely moves the weights’ quantisation error: the landscape within the subgroup is flat. The lower end of ViT-B’s range (0.76) marks cells where even the worst sampled rotation beat the identity —the only hint of structure, which Q2 picks up.

Figure 1 plots the distribution of e per regime —identity, shared orthogonal, per-head orthogonal, general GL— with the identity’s median marked as a horizontal reference line: the first three boxes overlap on that line in both models. The flatness is visible at a glance before reading a single figure, and the contrast with the fourth box (general GL, section 3.4) is immediate.

3.2 The trained gauge already sits by the norm-product floor (Q2)

The ratio between the identity’s error and the best sampled point under any regime lands on “close” in 768 of 768 Pythia cells (median $\times 1.02$, maximum $\times 1.10$) and in 237 of 288 ViT-B cells (median $\times 1.04$). The remaining 51 land in between, with a maximum of $\times 1.44$ and none in “far”. The reading is consistent with Q1: if the orbit is flat, the identity cannot be far from anything. Training does not optimise quantisability —nothing in its objective asks for it— but at these scales it does not need to either: the margin an optimal rotation could recover over the weights is a single-digit percentage, with the moderate exception of a fraction of the fine-tuned ViT’s heads (up to $\times 1.44$), insufficient to cross any threshold. The ratio compares the identity against the *best* sampled point, and corollary 2 closes the other half of the question: the minimum of the

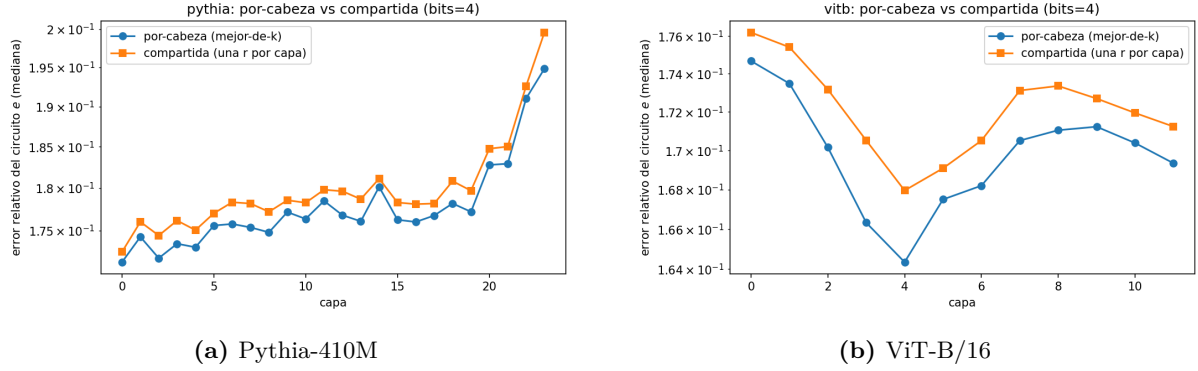


Figure 2: Per-layer median error of the best-of- K per-head versus the best shared rotation, a single one per layer (bits=4). The curves run parallel, with a gap of a little over one per cent across the 24 and 12 layers respectively.

norm product over the entire orbit, reachable and computable, sits 1.5 per cent away on Pythia and 1.1 per cent away on ViT-B. The trained gauge is not close to the best of thirty-two samples, it is close to the norm-product optimum.

3.3 Per-head freedom buys very little (Q3)

The shared regime admits a single $\mathbf{R}$ per layer, and that is how it is evaluated: of the $K = 32$ shared rotations, the one minimising the mean error over that layer’s heads is selected—the aggregate objective a rotational method would optimise—and compared head by head against the best-of- K per-head. The median relative improvement is 1.32 % on Pythia and 1.38 % on ViT-B, with 10th and 90th percentiles in $[0.13, 2.46]$ and $[0.15, 2.62]$: “null” by the preregistered 5 % threshold. On the weight side, sharing rotation across heads—SpinQuant’s restriction—is validated: the additional freedom the orbit allows per head buys a little over one per cent of circuit error, well below the threshold the design fixed as material.

Implementation correction. The first version of this reading took, within the shared arm, the per-head minimum over the K samples. Each head ended up choosing a different index, and the arm stopped being shared: the contrast compared per-head freedom against per-head freedom and gave, by construction, an improvement of -0.03% and -0.06% . Question Q3 and its threshold are the preregistered ones. What is corrected is the implementation of “shared”. The published code keeps both columns so that the correction is auditable.

The limitation of this contrast was preregistered: comparing random best-of- K ’s is not comparing optima. The local refinement over the orthogonal manifold was made conditional on Q1/Q2 revealing structure worth refining. With the flatness measured, it does not proceed and was not run.

Figure 2 plots the per-layer median of both curves—per-head and shared—. They run parallel in both models, separated by the same gap of a little over one per cent in every layer, with no layer where the gap widens. Q3’s verdict holds layer by layer and not only in the aggregate: the per-head advantage exists, it is uniform and it is small.

3.4 The large- p region is toxic—and why

In the sampled GL family, the error grows monotonically and severely with gauge strength. In int4, the median of e goes from 0.19 (scale 32, mild gauge, the identity’s own 0.19 to two decimals) to 1.10 (scale 8) and 2.85 (scale 2) on Pythia—0.19, 1.08, and 2.72 on ViT-B—, with maxima of 3.3×10^5 and 9.5×10^3 respectively. In int8 the pattern is identical one order of

magnitude below. It is the quantitative counterpart of the fp64 requirement: the phenomenon avoided there as numerical contamination appears here measured as a property of the space.

The mechanism admits a closed form. Under the gauge the factors become $\mathbf{W}'_v = \mathbf{R}^\top \mathbf{W}_v$ and $\mathbf{W}'_O = \mathbf{W}_O \mathbf{R}^{-\top}$, and the circuit recomposed after quantising is $\hat{\mathbf{M}} = \mathbf{M} + \Delta \mathbf{W}_O \mathbf{W}'_v + \mathbf{W}'_O \Delta \mathbf{W}_v + \Delta \mathbf{W}_O \Delta \mathbf{W}_v$. The quantiser used here bounds its own relative error, and that bound does not depend on the weights' scale.

Lemma 1 (Relative error of per-row RTN). *Let $\mathbf{W} \in \mathbb{R}^{m \times n}$ and let $\hat{\mathbf{W}}$ be its symmetric per-row RTN quantisation at b bits, that is $\hat{\mathbf{w}} = s \text{round}(\mathbf{w}/s)$ with $s = \|\mathbf{w}\|_\infty / q_{\max}$ and $q_{\max} = 2^{b-1} - 1$ for each non-zero row $\mathbf{w}$. Then*

$$\|\Delta \mathbf{W}\|_F \leq \frac{\sqrt{n}}{2q_{\max}} \|\mathbf{W}\|_F, \quad (1)$$

and the factor depends only on the row length and the bit width.

Proof. For one row, rounding to a grid of step s gives $|\Delta w_i| \leq s/2$ componentwise, and the clamp never acts because $|w_i/s| \leq q_{\max}$ by construction of s . Then $\|\Delta \mathbf{w}\|_2 \leq \sqrt{n} s/2 = \sqrt{n} \|\mathbf{w}\|_\infty / (2q_{\max}) \leq \sqrt{n} \|\mathbf{w}\|_2 / (2q_{\max})$, using $\|\cdot\|_\infty \leq \|\cdot\|_2$. A zero row has $\Delta \mathbf{w} = \mathbf{0}$. Summing squares over rows, $\|\Delta \mathbf{W}\|_F^2 \leq (\sqrt{n}/(2q_{\max}))^2 \|\mathbf{W}\|_F^2$. $\square$

Lemma 1 supplies the hypothesis the next proposition needs, with $\delta = \sqrt{n}/(2q_{\max})$ fixed within each factor. That suffices for a bound, and not only for an estimate.

Proposition 1 (Norm-product bound). *Let $\mathbf{W}_v$ and $\mathbf{W}_O$ have full rank, let $\mathbf{M} = \mathbf{W}_O \mathbf{W}_v$ and let $\mathbf{R} \in GL(d_h)$. If quantisation satisfies $\|\Delta \mathbf{W}'_v\|_F \leq \delta_v \|\mathbf{W}'_v\|_F$ and $\|\Delta \mathbf{W}'_O\|_F \leq \delta_O \|\mathbf{W}'_O\|_F$, then*

$$e(\mathbf{R}) = \frac{\|\hat{\mathbf{M}} - \mathbf{M}\|_F}{\|\mathbf{M}\|_F} \leq (\delta_v + \delta_O + \delta_v \delta_O) g(\mathbf{R}), \quad g(\mathbf{R}) = \frac{\|\mathbf{W}_O \mathbf{R}^{-\top}\|_F \|\mathbf{R}^\top \mathbf{W}_v\|_F}{\|\mathbf{M}\|_F}. \quad (2)$$

With δ_v and δ_O taken as uniform bounds over the family of gauges considered, the whole explicit dependence on the gauge sits in g .

Proof. The decomposition of $\hat{\mathbf{M}} - \mathbf{M}$ above, the triangle inequality and $\|\mathbf{AB}\|_F \leq \|\mathbf{A}\|_F \|\mathbf{B}\|_F$ give $\|\hat{\mathbf{M}} - \mathbf{M}\|_F \leq (\delta_O + \delta_v + \delta_O \delta_v) \|\mathbf{W}'_O\|_F \|\mathbf{W}'_v\|_F$. Divide by $\|\mathbf{M}\|_F$, which the gauge leaves fixed. $\square$

Normalising by its value at the identity the one factor of eq. (2) that depends on the gauge leaves the variable the rest of the section measures,

$$p(\mathbf{R}) \equiv \frac{g(\mathbf{R})}{g(\mathbf{I})} = \frac{\|\mathbf{W}_O \mathbf{R}^{-\top}\|_F \|\mathbf{R}^\top \mathbf{W}_v\|_F}{\|\mathbf{W}_O\|_F \|\mathbf{W}_v\|_F}. \quad (3)$$

Two bounds do not divide: from $e(\mathbf{R}) \leq C g(\mathbf{R})$ and $e(\mathbf{I}) \leq C g(\mathbf{I})$ it does not follow that $e/e_0 \leq p$. If the effective quantisation factor and the bound's slack vary little across gauges, the bound suggests $e(\mathbf{R})/e(\mathbf{I}) \simeq p(\mathbf{R})$, exponent one on paper and with nothing fitted. That is the hypothesis the rest of the section contrasts, and not a corollary of proposition 1. Two immediate consequences name the two ends of the design space.

Corollary 1 (The conformal orthogonal class is neutral for the bound). *For $\mathbf{R} = c\mathbf{Q}$ with $c > 0$ and $\mathbf{Q}$ orthogonal, $g(c\mathbf{Q}) = g(\mathbf{I})$, that is $p \equiv 1$ over the whole class, for every pair $(\mathbf{W}_v, \mathbf{W}_O)$.*

Proof. The Frobenius norm is orthogonally invariant, so $\|(c\mathbf{Q})^\top \mathbf{W}_v\|_F = c \|\mathbf{W}_v\|_F$ and $\|\mathbf{W}_O (c\mathbf{Q})^{-\top}\|_F = c^{-1} \|\mathbf{W}_O\|_F$. The factor c cancels in the product. $\square$

Corollary 2 (The minimisers of the bound). $\min_{\mathbf{R} \in GL(d_h)} g(\mathbf{R}) = \|\mathbf{M}\|_*/\|\mathbf{M}\|_F$. The set of minimisers is the pair’s balanced family —the one the previous work [1] characterises as the minimiser of $\|\mathbf{R}^\top \mathbf{W}_v\|_F^2 + \|\mathbf{W}_O \mathbf{R}^{-\top}\|_F^2$ — composed with the reciprocal rescalings $(\mathbf{A}, \mathbf{B}) \rightarrow (c\mathbf{A}, c^{-1}\mathbf{B})$, $c > 0$, and nothing else. It is therefore a single orbit of the conformal orthogonal class of corollary 1, which reappears here as the group carrying one minimiser to another.

Proof. For every factorisation $\mathbf{M} = \mathbf{AB}$, Hölder’s inequality for Schatten norms with $1 = \frac{1}{2} + \frac{1}{2}$ gives $\|\mathbf{M}\|_* \leq \|\mathbf{A}\|_F \|\mathbf{B}\|_F$. With $\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ of rank d_h , the balanced pair $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}^{1/2}$, $\mathbf{B} = \mathbf{\Sigma}^{1/2}\mathbf{V}^\top$ satisfies $\|\mathbf{A}\|_F \|\mathbf{B}\|_F = \text{tr } \mathbf{\Sigma} = \|\mathbf{M}\|_*$ and attains equality. The product is invariant under $(\mathbf{A}, \mathbf{B}) \rightarrow (c\mathbf{A}, c^{-1}\mathbf{B})$, so every reciprocal rescaling of a minimiser is a minimiser, even though it breaks the balance for $c \neq 1$.

Conversely, let $(\mathbf{A}, \mathbf{B})$ be any minimiser and $c^* = (\|\mathbf{B}\|_F/\|\mathbf{A}\|_F)^{1/2}$. The pair $(c^*\mathbf{A}, (c^*)^{-1}\mathbf{B})$ preserves the product and remains a minimiser, and both its norms now equal $\|\mathbf{M}\|_*^{1/2}$. Its sum of squares is $2\|\mathbf{M}\|_*$, the minimum the balanced-point lemma [1] characterises, so that pair is balanced. Every minimiser of the product is a reciprocal rescaling of a balanced one. $\square$

The design space’s border is thereby written down, and it is not the one between orthogonal and non-orthogonal. Corollary 1 marks the conformal orthogonal class $\{c\mathbf{Q}\}$ as neutral for the bound—the same class the previous work identifies as the only harmless one for $\mathbf{v}_1(\mathbf{W}_O)$, and of measure zero in $GL(d_h)$ by dimension, since $1 + d_h(d_h - 1)/2 < d_h^2$ for $d_h > 1$ —, and corollary 2 puts a floor on how much anisotropy can buy. From that family of minimisers a balanced representative is constructed, characterised within each reciprocal-rescaling family by having its two norms equal. That floor is measured: p at the balanced point is 0.985 on Pythia and 0.989 on ViT-B (median over heads), so the trained gauge sits within 1.5 per cent and 1.1 per cent of the optimum of the entire orbit in the variable of eq. (3). The design space is asymmetric within itself: upwards it reaches $\times 10^5$, while the improvement available below the trained gauge is about one per cent.

Conditioning is not enough. The prediction was contrasted with a sweep of gauges of controlled conditioning—the SVD of a Gaussian with its spectrum replaced by a geometric one between 1 and κ —, over nineteen grid points and with the main sweep’s counting gates: 642 048 additional measurements. The reading rule was fixed in writing before running, and on the condition number the verdict is **trend**, not law: at int4 the signal band spans only four grid points, and at int8, despite an R^2 of 0.9997 and an exponent that agrees across the two architectures (0.884 in both), the residuals retain systematic curvature above the threshold fixed. The preregistered consistency check fails as well: at matched κ , the controlled and the original family do not agree (median ratio 0.69–0.72 above $\kappa \approx 30$, with 5 of 48 original gauges inside the $[p_5, p_{95}]$ of their controlled counterparts). The condition number is not a sufficient statistic: two gauges equally ill-conditioned and of different spectral shape quantise differently.

The product law (declared post-hoc addition). The variable that does collapse the two families is that of eq. (3). Over 743 424 measurements from both families, two models and two bit widths, the log-log fit of e/e_0 against p gives exponent 0.9979 with $R^2 = 0.9993$ and a between-family separation of 0.003 in log. Against κ , exponent 0.714 and separation 0.167 (fig. 4). The derivation and its tolerance were written before this contrast, but the variable was chosen after seeing the κ one fail: the result enters as a declared post-hoc addition, just like the post-hoc forwards of section 3.5. What sustains it is the paired test—each original gauge against sixteen controlled ones at the same κ and on the same weights—, because in the pooled fit the families do not cover the same range of p and their separation would mix difference with range mismatch.

The tail, explained. With κ fixed the dispersion within a cell almost vanishes: the worst maximum-to-median ratio is $1.27\times$ on Pythia and $1.64\times$ on ViT-B. In the original family, at

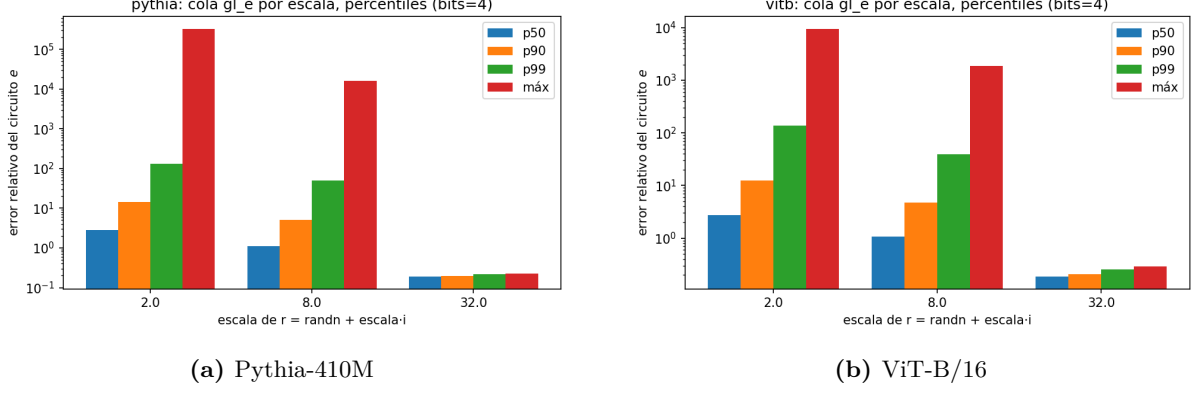


Figure 3: Percentiles (p50/p90/p99/max) of the circuit error for the general GL regime, per scale (bits=4). The maximum spikes above the p99 at low scale.

fixed scale, that ratio is in the thousands — $8\,224\times$ on Pythia at scale 2— and matches almost exactly that of κ within the same scale ($8\,181\times$). The catastrophic tail of fig. 3 is not a separate phenomenon: it is the dispersion of conditioning within each scale, transmitted by a law of exponent one. And in the orthogonal subgroup $p = 1$ exactly, so the factor of eq. (3) does not inflate: the boundary separating the flat region from the unquantisable zone is now derived for the bound’s factor, and the experiments show that it tracks the transition to the high-error region.

Figure 3 does not use a box with hidden outliers —with a median of 2.85 and a maximum of 3.3×10^5 on Pythia at scale 2, hiding the tail would hide the headline—. It instead reports the p50/p90/p99 percentiles and the maximum, explicit, per scale. The maximum spikes far above even the p99: the tail is of the extreme, not of the body of the distribution.

The balanced point: the law tested from the deflating side. The fit of eq. (3) rests on families that inflate p . Corollary 2 lets one construct explicitly a representative with $p < 1$, the balanced point, by SVD of the circuit rather than sampling it, so the prediction can be contrasted outside the fitted range. Two readings were fixed in writing before running: the balanced point *improves* if $e_{\text{bal}} < e_{\text{id}}$ in at least 90 per cent of the cells, and the law *extrapolates* if the median of $(e_{\text{bal}}/e_{\text{id}})/p_{\text{bal}}$ falls in $[0.90, 1.10]$.

The result separates the two. The law extrapolates by the fixed threshold —median 1.042 on Pythia and 1.079 on ViT-B— and the balanced point does not improve: the measured ratio is 1.027 and 1.066 in int4, with only 7.8 per cent and 2.1 per cent of the cells below the trained gauge. The deviation has one sign across both models and both bit widths, and it is the price of going from the bound to the approximation: $e/e_0 \simeq p$ treats the effective quantisation factor as roughly constant across gauges, and over 1.5 per cent of travel that assumption no longer holds.

The measurement matters for another reason. The balanced point is a *non-orthogonal* gauge — $d(\mathbf{R}) = 0.089$ on Pythia and 0.076 on ViT-B, with $\text{cond}(\mathbf{R})$ of 1.61 and 1.44—, and its error stays at $\times 1.03$ and $\times 1.07$ against the median $\times 15$ and the maximum $\times 10^5$ of the sampled general orbit. Leaving the orthogonal subgroup does not degrade by itself: what governs is p , and an anisotropic gauge with $p \simeq 1$ quantises like a rotation. Here $d(\mathbf{R})$ is the standard deviation of the singular values divided by their norm, the distance to $\{c\mathbf{Q}\}$ that the previous work [1] uses as the gauge’s strength dial, and it is exactly zero on the conformal orthogonal class.

3.5 End-to-end validation

Seven forwards of ViT-B over the complete clean val of ImageNet-100 (5000 images). Five with $(\mathbf{W}_v, \mathbf{W}_O)$ in int4 —identity, best and worst sampled orthogonal gauge, balanced point and a general gauge at scale 2— and two unquantised controls, the identity and that same general

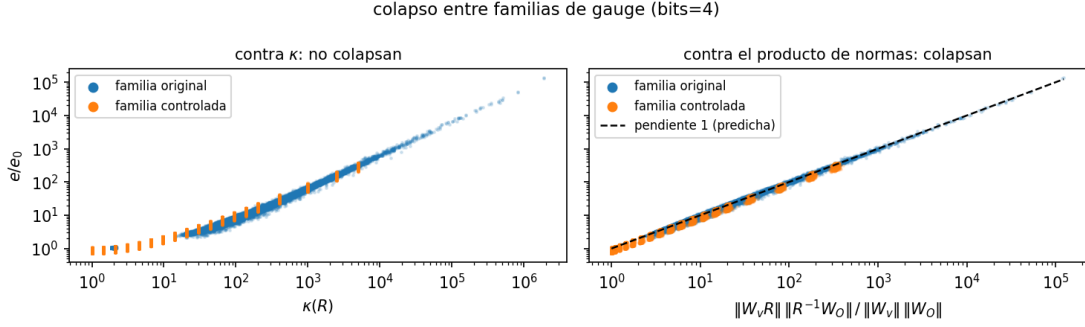


Figure 4: Collapse between gauge families (bits=4). Against κ (left) they are offset; against the product of eq. (3) (right) they fall on the slope-one line, drawn unfitted. Declared post-hoc addition.

gauge. The sampled gauges are reconstructed exactly by replaying the generator sequence the sweep already consumed, with no need to persist each rotation, and the reconstruction was verified against the sweep’s csv on three control cells (difference $< 10^{-9}$). Every top-1 is in table 1. The conditions are evaluated over the same images and in the same order, so each contrast carries its paired test: exact McNemar over the discordant pairs and a 95 per cent interval by image bootstrap (10 000 resamples, fixed seed).

Implementation correction. The gauge is applied to the pair and to the value bias. Since $\mathbf{v} = \mathbf{x}\mathbf{W}_v^\top + \mathbf{b}_v$, compensating $\mathbf{W}_O$ requires $\mathbf{b}_v \rightarrow \mathbf{R}^\top \mathbf{b}_v$, and an earlier version of this table left the bias untouched: over an unquantised batch, an orthogonal gauge then moved the output by 57 per cent in relative terms and only 12 per cent of the top-1 predictions agreed. The forward was measuring a different model, and the table credited the identity with an advantage of 2.8 points that was that bias. A gate now verifies the forward’s invariance *before* quantising, condition by condition: 4×10^{-7} on the three well conditioned gauges and 1.2×10^{-5} on the general one at scale 2.

Preregistered contrast: orthogonal indistinguishability, confirmed. Best and worst sampled orthogonal gauge give 0.9306 and 0.9298, and the identity 0.9308: a range of 0.10 percentage points across the three. The three paired contrasts are indistinguishable, with McNemar at 0.68, 1.00 and 0.52 and every interval containing zero. The identity falls inside the plateau rather than outside it, which is Q1’s updated prediction carried into real accuracy.

One nuance the predicted-class agreement reveals and top-1 hides: across those quantised conditions the prediction agrees on 98.4 to 98.9 per cent of the images, not on all of them. Quantisation reshuffles individual cases even when aggregate accuracy does not move. Indistinguishable in accuracy, therefore, and not identical in prediction.

Declared post-hoc addition: the balanced point. The gauge corollary 2 constructs —non-orthogonal, with $\text{cond}(\mathbf{R}) = 1.44$ on this model— gives 0.9306, within 0.02 points of the identity and with McNemar at 1.00. Leaving the orthogonal subgroup on the side the bound predicts as good costs no accuracy. The claim that “the orthogonal restriction is necessary” is refuted by a constructed counterexample, and corollary 1 holds in its place: what is dangerous is p far from one, and not the absence of orthogonality.

Declared post-hoc addition: the GL collapse and its control. The general gauge at scale 2 is measured twice, with and without quantisation. Unquantised it gives 0.9314, the same top-1 as the identity in fp32, and the paired contrast goes further than top-1: **the predicted class agrees on all 5000 images**, without a single discrepancy. The network predicts exactly the

Table 1: End-to-end validation on ViT-B, clean val of ImageNet-100 (5000 images). The invariance column is the relative deviation of the forward under the gauge *before* quantising. The first six contrasts are indistinguishable by paired McNemar. The seventh, identity against the general gauge with both in int4, is the only distinct one, and with the sixth it isolates the cause: equal unquantised, different quantised.

Condition	Bits	Top-1	Invariance	Status
Identity	—	0.9314	—	control
Identity	4	0.9308	—	reference
Best sampled orthogonal	4	0.9306	4.0×10^{-7}	preregistered
Worst sampled orthogonal	4	0.9298	4.0×10^{-7}	preregistered
Balanced point	4	0.9306	4.2×10^{-7}	declared post-hoc
GL scale 2	—	0.9314	1.2×10^{-5}	control
GL scale 2	4	0.0082	1.2×10^{-5}	declared post-hoc

same thing, image by image, under a change of coordinates of high condition number. Quantised to int4 it gives 0.0082, below the 1.00 per cent that nominal chance spreads over 100 classes, a drop of 92.26 points. The control isolates the cause: the gauge does not touch the function and quantisation destroys it. Gauge equivalence is exact in real arithmetic and does not survive the grid.

4 Discussion

Taken together. The three results compose an asymmetry, and the closed form names its border. Within the orthogonal subgroup, neither the point (Q1, Q2) nor per-head granularity (Q3) materially move the weights’ quantisation error, and section 3.5 confirms they do not distinguish the model’s real accuracy from each other either. In the large- p region, degradation reaches collapse.

The border, however, does not separate orthogonal from non-orthogonal. Corollary 1 leaves the conformal orthogonal class $\{c\mathbf{Q}\}$ exactly neutral for the bound, and the balanced point —anisotropic, with $d(\mathbf{R}) \simeq 0.08$ — quantises like a rotation. What separates the plateau from the unquantisable region is p , and orthogonality is the cheap sufficient condition that pins it at 1 without computing it. For rotational quantisation the implication can then be stated without excess: the orthogonal restriction is no implementation convenience, because it guarantees $p = 1$ by construction, and the sharing restriction leaves almost nothing on the table on the weight side. The bound predicts the same risk for an anisotropy with $p \simeq 1$, and the balanced point is the case section 3.5 validates end-to-end. Evaluating p costs two Frobenius norms.

Practical implications. Four design guidelines follow from the characterisation, each from its result. First, from Q3: implementing per-head rotations in a rotational method buys very little on the weights side —SpinQuant’s shared restriction is almost free at these scales, and multiplying the number of stored or learned rotations by H offers little return at these scales—. Second, from Q1–Q2, the balanced point and the end-to-end validation: do not spend search budget optimising the gauge for weight error. The landscape is flat and corollary 2 bounds how much the *norm-product factor* can improve over the whole orbit from the trained gauge: 1.5 per cent on Pythia and 1.1 per cent on ViT-B. That bound does not carry over to the RTN error, and indeed the constructed minimiser does not realise it. The two together give an empirical recommendation rather than a theorem: at these scales there is little weight-side return in spending search budget on the gauge. The target of rotational optimisation should be the activation channel, where those methods report their gains. Third, from the product law: any non-orthogonal reparametrisation of the value-output sector —model surgery, merging,

gauge fixing— yields functionally identical checkpoints that may be unquantisable, and p gives a data-free estimate of their circuit-error inflation risk before quantising. It suffices to evaluate p in eq. (3) on the weights and the transformation: two Frobenius norms and one $d_h \times d_h$ inversion, with no data, no forwards and nothing quantised. A pipeline receiving third-party transformed checkpoints can reject them, undo the transformation, or budget the loss with that figure in hand. With the limit the third caveat imposes: p is the factor of a bound and not an exact prediction, and its measured deviation has one sign. Fourth, from corollary 1: a transformation of the value-output sector with a small anisotropic part carries a low risk of circuit-error inflation under this bound, and the cheap test is p rather than orthogonality. Fixing the gauge by QR, balancing the factors or rescaling heads are distinct operations for $\mathbf{v}_1(\mathbf{W}_O)$ and equivalent for quantisation as long as p stays near one.

First caveat: weights, not activations. This design measures the weight side of the circuit. The gains QuaRot and SpinQuant report come mostly from eliminating *activation* outliers, a channel this work does not touch by construction. The result does not say the rotations do not work; it says, by elimination of the other channel, that their benefit does not reside in the orbit’s weight landscape—a delimitation consistent with the attribution those same works make of their improvements.

Second caveat: scale. Pythia-410M and a fine-tuned ViT-B are small models, and weight outlier phenomena intensify with scale. “At these scales” is part of the claim, not fine print: the flatness of Q1–Q3 might not persist in 7B-plus models, and that check—cheap, the sweep is pure inference—is the natural extension that would decide whether the characterisation generalises.

Third caveat: from the bound to the approximation. Proposition 1 bounds with δ_v and δ_O fixed; the step to $e/e_0 \simeq p$ further treats them as roughly constant across gauges. Near the floor that assumption breaks, because the relative error RTN makes per row depends on the row’s shape and the gauge changes it. Over the families that inflate p the fit gives exponent 0.9979; at the balanced point, the only measured regime with $p < 1$, the prediction falls short by 4 per cent on Pythia and 8 per cent on ViT-B, always in the same direction. p detects the orbit’s dangerous region with room to spare and loses resolution among representatives already close to the floor. As a forecast of absolute error from the deflating side, it is an optimistic ceiling. We do not settle the cause of that bias—the per-row scale RTN picks depends on the row’s shape, and the gauge changes it—and leave it stated as a limit of the bound, not as noise.

Link to the program. The orbit as a design space has two faces. The compression one is characterised here: level over the neutral class and its neighbours, destructive when p grows. The functional one—whether the orbit and its deformations are a lever for inducing functional diversity across heads—is the object of the component-attribution program that continues [1], where the question is addressed by intervention rather than sampling.

Reproducibility

Code for the sweeps, the complete CSVs (170 016 rows for the regime sweep, 642 048 for the controlled-conditioning one, 102 432 for the re-sweep of the original family with norms and 1 056 for the balanced point), reading tables, and figure scripts at <https://github.com/mmunozp1/gauge-orbit-quantization>. The sweep is reproducible by global seed. Any sampled gauge can be reconstructed exactly by replaying the generator sequence, without persisting the matrices. The end-to-end validation stores, for every condition, the per-image correctness and the predicted class, from which the paired contrast, its intervals and the argmax agreement are computed.

Every gauge is applied to the pair and to the value bias, and a gate verifies the forward’s invariance before quantising, condition by condition.

An interactive demo accompanies the code at <https://huggingface.co/spaces/ManPla/gauge-orbit-quantization-demo>. Pick a point of the orbit and it measures p , $d(\mathbf{R})$ and the circuit error with and without quantisation, live. It reimplements nothing: it runs the same `src/` that produced the tables on a light carrier of the value-output sector, and a gate certifies it cell by cell against the published sweeps, with a worst difference of 3×10^{-15} .

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